



Aerodynamic Interactions among Three Identical Wings in Close Proximity

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The compactness requirements for modern Unmanned Aerial Vehicles (UAVs) and Private Air Vehicles (PAVs) opens the door for new innovative designs with smaller footprints. Some of the proposed UAV and PAV designs utilize multiple lifting surfaces or rotors in close proximity increasing the need for strategic placement of the wings within close proximity. The interactions among three wings in close proximity were experimentally quantified as a function of stagger, gap, and décalage angle. All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) on three identical Clark-Y AR 2 semi-span wings. Changes in aerodynamic force coefficients of the individual wings were used to establish the proximity effects at different stagger and gap locations. Beneficial and detrimental combinations of stagger, gap, and décalage angle were selected for optimization of multiple wing aircraft concepts.

I. Nomenclature

St	= Stagger (Lateral Spacing)	$C_{L\alpha}$	= Lift Curve Slope
G	= Gap (Longitudinal Spacing)	C_D	= Drag Coefficient; $C_D = \frac{Drag}{(1/2)*\rho*U^2*S}$
α	= Geometric Angle of Attack	L/D	= Lift to Drag Ratio
$\alpha_{L=0}$	= Zero Lift Angle of Attack		
c	= Chord		
b	= Wingspan		
AR	= Aspect Ratio; $AR = \frac{b^2}{S}$		
U_∞	= Freestream Velocity		
C_L	= Lift Coefficient; $C_L = \frac{Lift}{(1/2)*\rho*U^2*S}$		
		Subscripts	
		1	= Wing 1
		2	= Wing 2
		3	= Wing 3
		sys	= System

II. Introduction

Unmanned Aerial Vehicles (UAVs) and Private Air Vehicles (PAVs) have become the recent focus of the aerospace industry. While there are certain benefits and advantages of these vehicles from an operational standpoint, aircraft designers face new problems connected to size requirements. For a large number of UAVs and PAVs to be practical in more densely populated areas, the footprint of these designs must be reduced. The biggest factor in a fixed-wing aircraft's footprint is the wingspan and therefore the idea of replacing the conventional large span wing with multiple small span wings becomes increasingly enticing.

Multiple lifting surfaces and air taxis are not a new concept. While biplanes are the first designs to come to mind, other more exotic designs exist such as the Zerbe Air Sedan [1] from the 1920s to the modern electric Faradair BEHA

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[2] as shown in Figure 1. The Zerbe Air Sedan built by Jerome S. Zerbe featured four wings with equal stagger and gap between them. The Faradair BEHA is a hybrid-electric triple box wing aircraft designed to carry around 18 passengers.

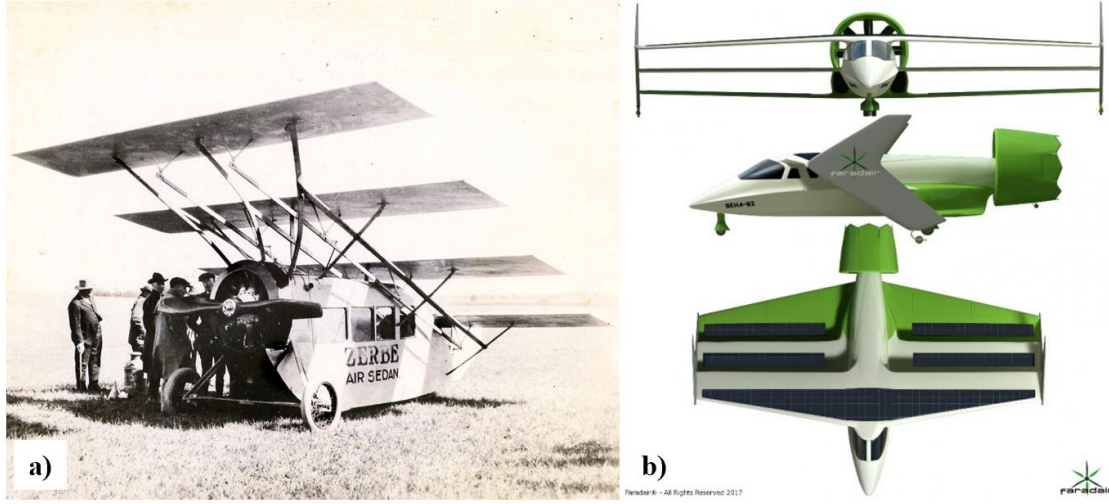


Fig. 1 Multiple Wing Concepts of a) Zerbe Air Sedan [1] b) Faradair BEHA [2]

Fundamental work on multi-plane or multi-element aerodynamics, specifically biplanes and triplanes, was published by Prandtl [3] and Munk [4] back in the 1920s. The studies derived theorems to predict the induced drag and lift distributions to attain favorable configurations. Prandtl [3] used an influence coefficient, σ , that can be found in tables as a function of gap and span to approximate the total induced drag on triplanes with Eq. (1). In Eq. (1), σ_1 represents the mutual influence on the top or bottom wing with the middle wing, σ_2 is the influence on the top and bottom wing, and x is the coefficient of distribution of wing loads. The variables σ and x are mathematically expressed in Eq. (2) and Eq. (3) respectively. The assumptions made for the following theory were equal span and gap spacing between the three wings as well as that the upper and lower wing generate the same amount of lift at the minimum drag condition.

$$D = \frac{L^2}{2\pi q b^2} [1 + \sigma_2 - 2x(1 + \sigma_2 - 2\sigma_1) + x^2(3 + \sigma_2 - 4\sigma_1)] \quad (1)$$

$$\sigma = f(G, b) \quad (2)$$

$$x = \frac{L_2}{(L_1 + L_2 + L_3)} \quad (3)$$

More recently, experimental studies were performed on similar research in the form of biplanes and multi-wing aircraft concepts at the University of Dayton. Kang et al. [5,6] studied the effects of gap and stagger on biplane configurations with endplates where downwash angle was shown to be an important factor in the performance and the reason for direction of gap and stagger having an effect. Multi-wing configurations were experimentally and numerically investigated in studies by Mongin et al. [7,8,9] for distributed lift concepts. It was found that the multi-wing configurations could achieve around 50% of the aerodynamic efficiency that an equivalent mono-wing could achieve with minimal differences between the different configurations.

Wing-wing interactions between two wings in close proximity was quantified by Jestus et al. [10]. The study confirmed that placing a wing directly upstream of another wing in close proximity can lead to a decrease in aerodynamic efficiency of the aft wing. A secondary wing along a diagonal of high gap and low stagger saw the least amount of performance decrement. System analysis of the two wings combined saw true biplane configuration with 0c stagger and 1c gap to have high performance at several instances of Wing 1 and Wing 2 angle of attack combinations.

Extending upon the study by Jestus et al. [10], this study looks to quantify the interactions among three identical wings as a function of proximity. Analyzing the changes in performance of each individual wing can allow inter-wing optimization for more complex multi-wing concepts.

III. Experimental Setup

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open-jet configuration. The tunnel has a 16:1 contraction ratio and a test section size of 76.2 cm by 76.2 cm by 137 cm with velocities ranging from 6.7 m/s to 36.7 m/s. The maximum turbulence intensity was measured to be 0.1% at 15 m/s by hot-wire.

The experiments were conducted with three Clark Y AR 2 semispan wings with a chord of 0.127 m and span of 0.254 m. These wings were placed at various stagger and gap locations as well as relative angles of attack (décalage). Wing 1 was fixed in the center of the XY plane, while Wing 2 and Wing 3 were grouped in either a pure 1 chord gap biplane or 1 chord stagger tandem wing configuration to move around Wing 1 in varying stagger and gap positions. Wing 1 was attached to a 6 component Analog ATI Gamma Net F/T sensor [11] which was mounted on top of a PDV PT-GD201 rotary stage [12] to adjust the angle of attack. Similarly Wing 2 was connected to a 6 component Digital ATI Gamma Net F/T sensor [11] on top of a PDV PT-GD201 rotary stage [12]. The final wing, Wing 3, measured the forces and torques with a 6 component ATI Mini40-E [13] and another PDV PT-GD201 rotary stage [12] to rotate angle of attack. Figure 2 shows a schematic of the experimental setup in the UD-LSWT.

The ATI Gamma Net F/T sensor was orientated with the normal and axial forces measured through the X and Y axis. The sensing ranges of the Gamma in the SI-65-5 Calibration of X and Y is 65 N with an uncertainty of 0.75% and 1.00% respectively along with the torque range of 5 Nm and uncertainty of between 1.00% to 1.50% depending on the axis. The ATI Mini40-E Net F/T sensor was orientated with the normal and axial forces measured through the X and Y axis. The sensing ranges of the Mini40 in the SI-40-2 Calibration of X and Y is 40 N with an uncertainty of 1.25% for both axes along with the torque range of 2 Nm and uncertainty of 1.25% to 1.50%. A sampling rate of 1000 Hz was used during data acquisition. The PDV GD-201 rotary stage instrument uncertainty in angle of attack was 0.01°.

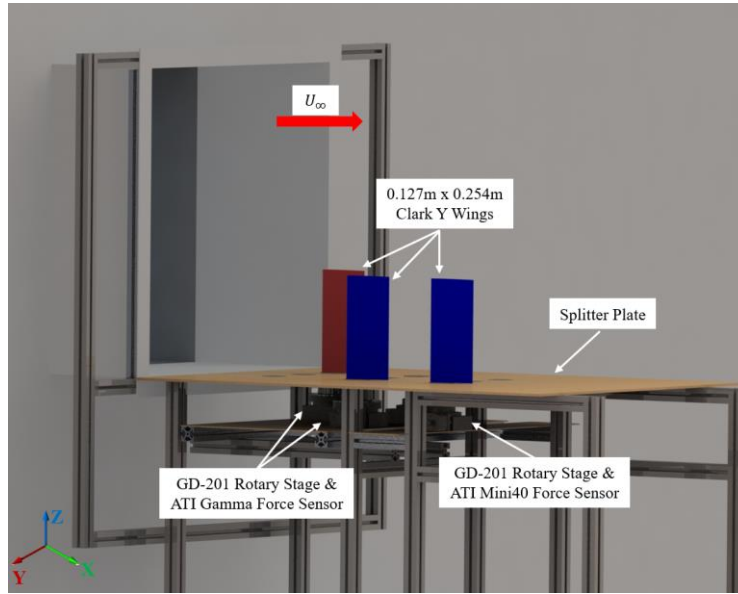


Fig. 2 Schematic of experimental setup in University of Dayton Low-Speed Wind Tunnel

The coordinate system was centered on the leading edge of Wing 1. The naming convention for the locations is that the x-coordinate represents the stagger distance in chord between Wing 1 and the centroid between Wing 2 and Wing 3 with the y-coordinate as the gap distance as annotated on the grid in Fig. 3. This is to remain consistent with biplane and previously published Distributed Lift nomenclature.

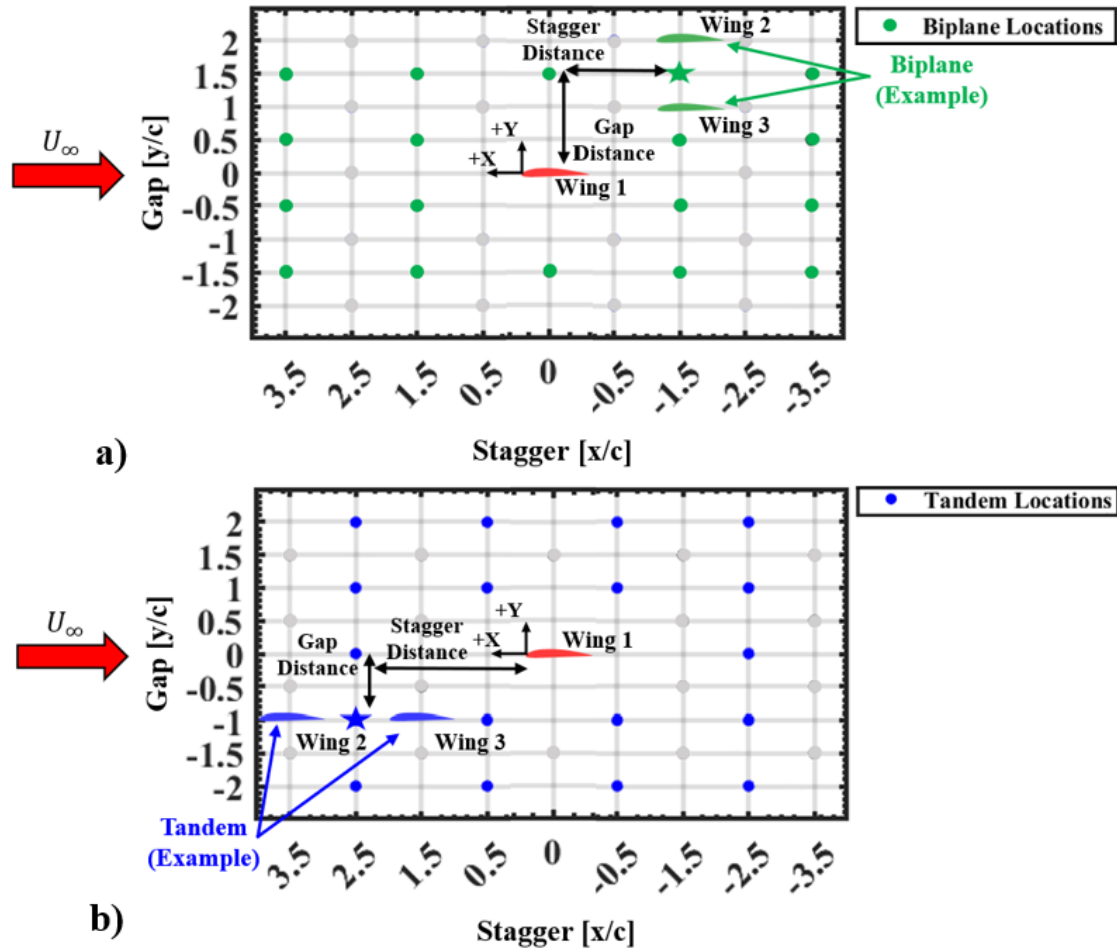


Fig. 3 Experimental nomenclature and Wing 2 and Wing 3 grid locations as a) Biplane and b) Tandem

Table 1 shows the test matrix of the experimental test campaign. The centroid of Wing 2 and Wing 3 in their respective biplane or tandem group was positioned at these stagger and gap locations at a fixed angle of attack while the Wing 1 angles (α_1) were changed from -20° to 20° . Then the biplane or tandem's angle was changed at the same stagger and gap location before a Wing 1 angle of attack sweep is performed again. The same procedure was used at all the different stagger and gap locations shown in Fig. 3. The three α_2 values were selected to capture zero lift angle ($\alpha_{L=0}$), intermediate attached flow, and the highest system efficiency Wing 2 angle based on the previous study by Jestus et al. [10]. The tests were performed at a baseline individual wing Reynolds number of 250,400 at a freestream velocity of 30 m/s.

Table 1 Experimental Test Matrix

Two Wing Group	Reynolds Number	Stagger Distances [x/c]	Gap Distances [y/c]	Wing 1 Alpha Range [°]	Wing 2 and Wing 3 Alpha Range [°]
(0c St, 1c G) Biplane	250,400	3.5, 1.5, 0, -1.5, -3.5	1.5, 0.5, -0.5, -1.5	-20 to 20	-3, 3, 8
(1c St, 0 G) Tandem	250,400	2.5, 0.5, -0.5, -2.5	2, 1, 0, -1, -2	-20 to 20	-3, 3, 8

IV. Results

The changes in performance of Wing 1, Wing 2, and Wing 3 in close proximity at different 2D spacing is discussed in this section. The 2D stagger and gap spacing with respect to the center wing are annotated in Fig 3.

To attain a comparable single wing baseline case, one of the Clark Y wings was tested separately at different positions in the spanwise direction and downstream of the wind tunnel inlet to ensure repeatability at various spatial locations in the test section. Figure 4 shows the averaged lift curve of the single wing from all the different tests along with the dashed lines representing the error bars. The baseline cases see negative stall around 10° and positive stall around 18° with a $C_{L_{Max}}$ of 1.1. The lift curve slope (C_{L_α}) in the linear region of the averaged curve is calculated to be 0.057 deg^{-1} which is around $C_{L_\alpha} = \pi\alpha$. This is consistent with low AR wing behavior. Using Helmbold's low aspect ratio equation [14] in Eq. (4), the aspect ratio corrected lift curve slope of the averaged baseline is calculated with C_{L_α} as 2π for thin airfoils and an effective AR of 2.8. The corrected lift curve slope was derived to be 0.0564 deg^{-1} and is also plotted in Fig. 4 for comparison.

$$C_{L_\alpha} = C_{l_\alpha} \frac{AR}{\left(\frac{C_{l_\alpha}}{\pi}\right) + \sqrt{\left(\frac{C_{l_\alpha}}{\pi}\right)^2 + AR^2}} \quad (4)$$

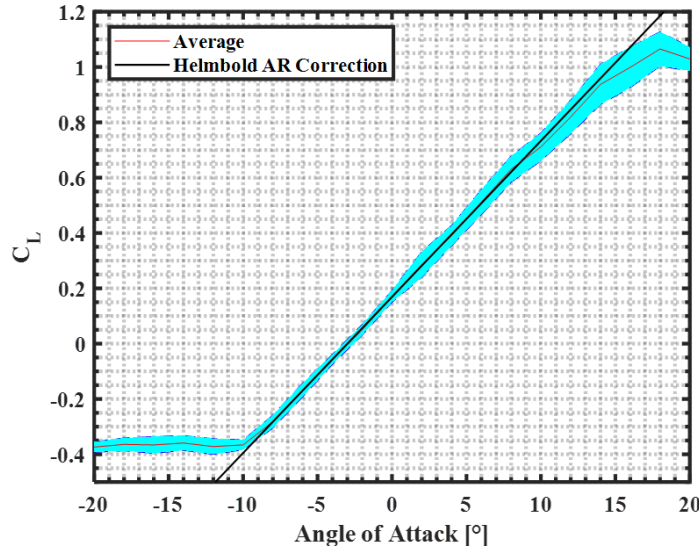


Fig. 4 Baseline AR 2 Clark Y Lift Curve

As mentioned above, the study by Jestus et al. [10] looked at 1 chord gap biplane and 1 chord stagger tandem configurations when Wing 1 was sweeping through an angle of attack range and Wing 2 was at a fixed angle of attack. Those results are revisited here to help establish the performance interactions on what will be referred to as the biplane and tandem groups for this study in isolation without a third wing.

Figure 5 presents the biplane configuration where a clear shift of the lift curve to the right occurs when Wing 2 was at 8° compared to minimal differences when Wing 2 was at -3° . The drag coefficient of Wing 1 did not appear to be as sensitive to Wing 2's angle of attack. In terms of Wing 2's performance, C_L had similar trends for both α_2 relative to the baseline curves whereas Wing 1 increased in angle of attack, Wing 2 would generate less lift. An increase in Wing 1 angle of attack also resulted in an increase in drag, but for Wing 2 at 8° , even this increasing trend never increased drag above the baseline. It can also be seen that Wing 1 and Wing 2 produce the same C_L when both wings are at -3° or 8° .

For the 1 chord stagger tandem configurations in Fig. 6, Wing 1 is only slightly affected by the presence of Wing 2 downstream. Wing 2 on the other hand suffers a large decrement in C_L as Wing 1 angle of attack increases. The C_D of Wing 2 at -3° also witnesses an increase in C_D at higher α_1 , but when at 8° there is a bell-shaped trend when C_D increases then drops towards the baseline's minimum as Wing 2's induced angle of attack decreases in the wake of Wing 1.

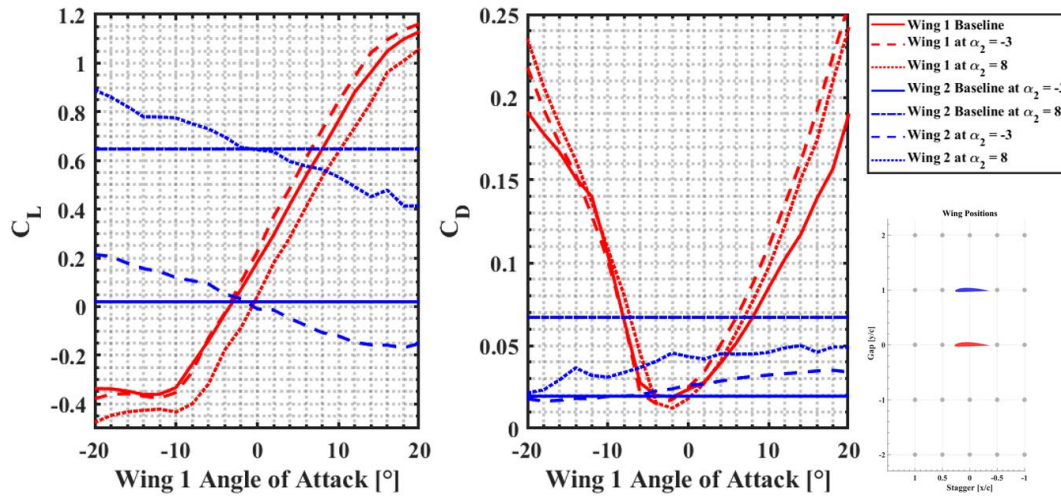


Fig. 5 Wing 1 and Wing 2 performance in isolated (0 chord Stagger, 1 chord Gap) Biplane configuration with Wing 2 at -3° and 8°

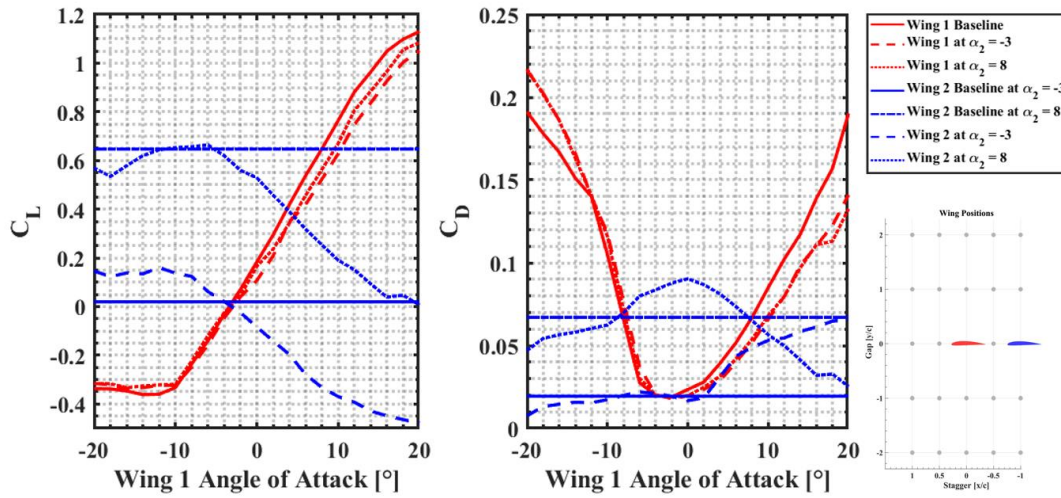


Fig. 6 Wing 1 and Wing 2 performance in isolated (1 chord Stagger, 0 chord Gap) Tandem configuration with Wing 2 at -3° and 8°

With both Wing 1 and Wing 2 at various independent angles of attack, several instances exist when both wings are at the same angle of attack. In the current study, Wing 2 and Wing 3 will always remain in a biplane or tandem configuration at the same angle of attack and therefore in order to achieve a baseline measurement that accounts for the wing-wing interactions, biplane and tandem system C_L at angles of attack of -10°, -3°, 8°, 13°, and 20° can be experimentally collected. Then, linear interpolation can be performed to obtain the C_L of the biplane or the tandem as single system at 3° angle of attack. Figure 7 shows this linear trend for both the system C_L of the biplane and tandem configurations. Both configurations show a deviation in the lift curve slope from the standalone wing with the tandem acting similar to an aspect ratio (AR) reduction from the biplane.

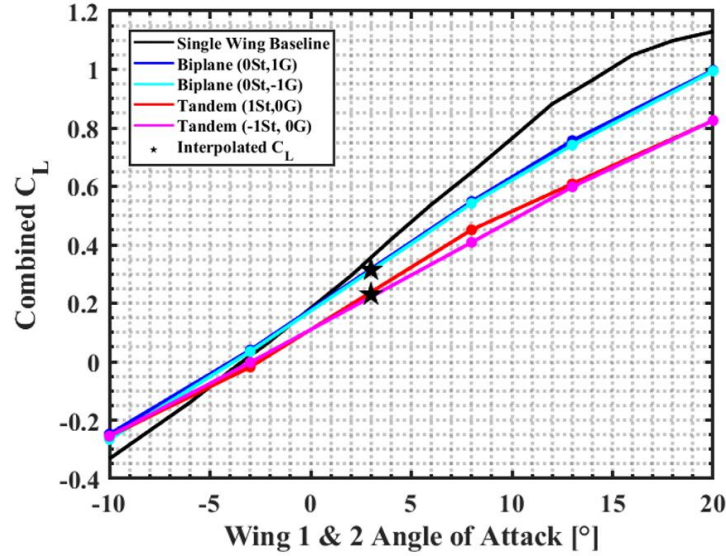


Fig. 7 Combined C_L of Two Wings at the same angle of attack in a Biplane or Tandem Configuration

A. Wing - Biplane Group Results

The standalone wing and the two-wing biplane at the same angle of attack established in Fig. 4 and Fig. 7 were used to compare the performance of the three wings in close proximity. In the reference frame of Wing 1, the measured forces were subtracted by the baseline so that blue in the contours represented an increment and red represented a decrement from the baseline standalone wing at that angle of attack. Looking at different Wing 1 and biplane group angle of attack combinations, Fig. 8 displays C_L changes in Wing 1 at each biplane group stagger and gap location. Wing 1 experiences mostly decrements, especially when the biplane group is upstream of it. The areas of best Wing 1 C_L , whether increment, no change, or least decrement, happen when the biplane group is downstream.

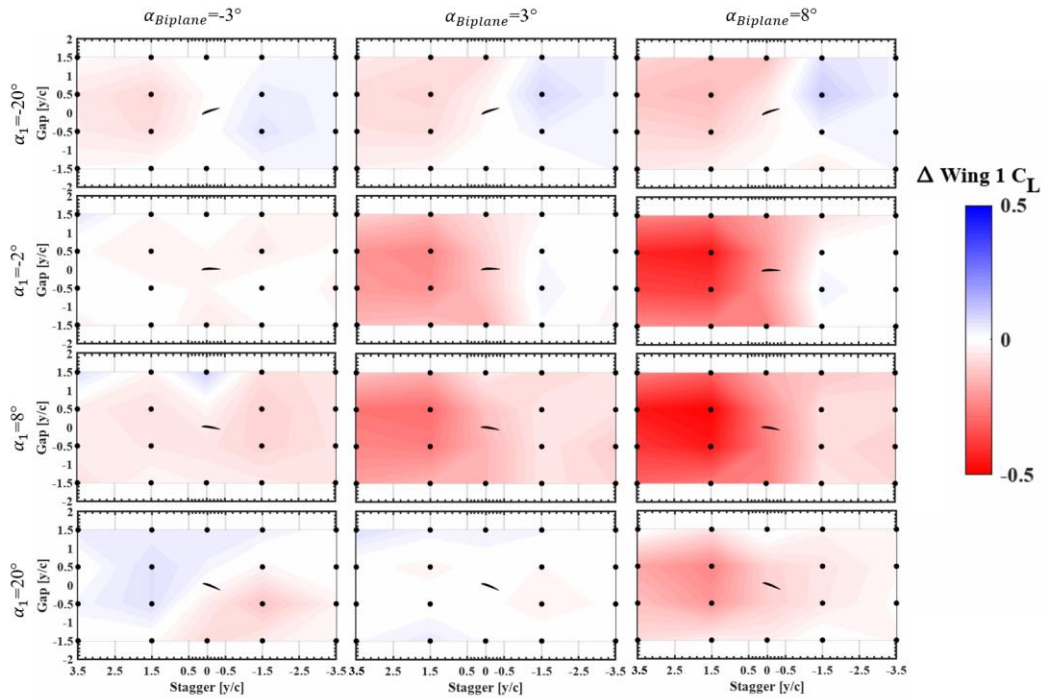


Fig. 8 Contours of Difference in Wing 1 C_L compared to Baseline C_L at different Wing Alpha Combinations

A similar approach can be used to now analyze the biplane group's performance at each grid point. For an effective comparison of the biplane group's performance changes due to the addition of a third wing, the difference between the biplane group's collective C_L while both the wings are at the same α was subtracted from the biplane group's collective C_L with the third wing as mathematically expressed in Eq. (5). Figure 9 of the contour array shows mostly decrements from the biplane group. Just as seen with Fig. 8 above, the decrements are the common trend with the strongest being when the measured wing is downstream of another wing. The contours on the top row when $\alpha_1 = -20^\circ$ does have locations where the biplane group experiences an increment in lift despite being downstream of Wing 1. The downstream location sees the strongest increment at the lowest angle of attack combination and the strongest decrement at the highest angle of attack combination out of all the locations possibly emphasizing the effects of downwash on the aft wings' performance.

$$\Delta C_{L_{Biplane}} = \left(\frac{L_2 + L_3}{\frac{1}{2} \rho U_\infty^2 (2 * S)} \right) - (C_{L_{Biplane \text{ baseline at } \alpha}}) \quad (5)$$

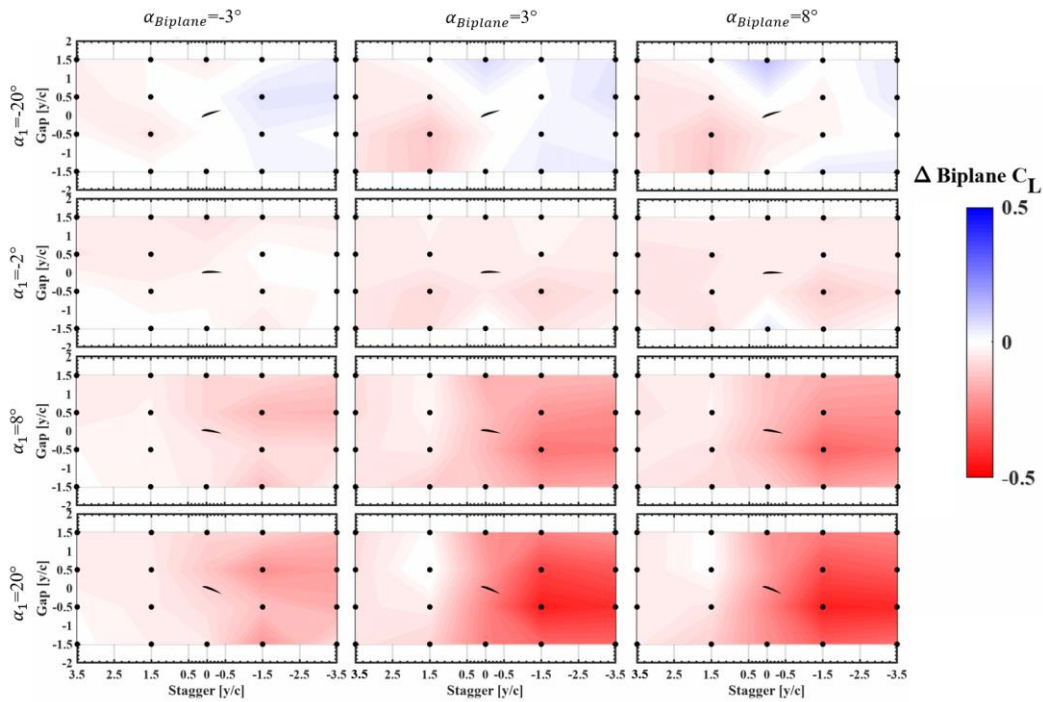


Fig. 9 Contours of Difference in Biplane Group C_L compared to Ideal Two Wing Baseline C_L at different Wing Alpha Combinations

B. Wing - Tandem Group Results

The wing interaction with the two-wing tandem group can be analyzed identical to the biplane analysis performed above. The effect of the tandem group on Wing 1's C_L is plotted in a contour array of different Wing 1 and tandem group angle of attack combinations in Fig. 10. All three wings when in the intermediate attached flow regime produce fully decrement contours as seen by $\alpha_1 = -2^\circ, 8^\circ$ and $\alpha_{Tandem} = 3^\circ, 8^\circ$. Increments can be observed on Wing 1 despite wings upstream when $\alpha_1 = 20^\circ$. This is due to stall being "delayed" as induced angle of attack is lower than the assumed α_1 because of the tandem group's wake.

The contours in Fig. 11 quantify the effect that the sweeping Wing 1 has on the tandem group. Again, the performance is normalized by subtracting the baseline of a two-wing tandem system with both wings at the same angle of attack. In the two wing studies, the tandem group appeared to suffer the largest decrements, but the addition of a third wing seems to be beneficial to Wing 2 and Wing 3 in the tandem group. Some of the instances of increment occur around a stagger of $0.5c$ which means that Wing 2 or Wing 3 in the tandem group is in a biplane configuration with Wing 1. When the two wings are downstream at their $\alpha_{L=0}$ and Wing 1 is in a negative angle of attack stall, this

increment trend is more dominant than a decrement. Once Wing 1 increases angle of attack to produce positive lift, the tandem wings experience a decrement when downstream. In Fig. 8 through Fig. 10, all three wings at $\alpha_{L=0}$ show the smallest deviation from the baseline, but in the tandem wing case of Fig. 11, this trend is taken to an extreme where throughout the whole testing grid there are only a few instances of any change in coefficient of lift.

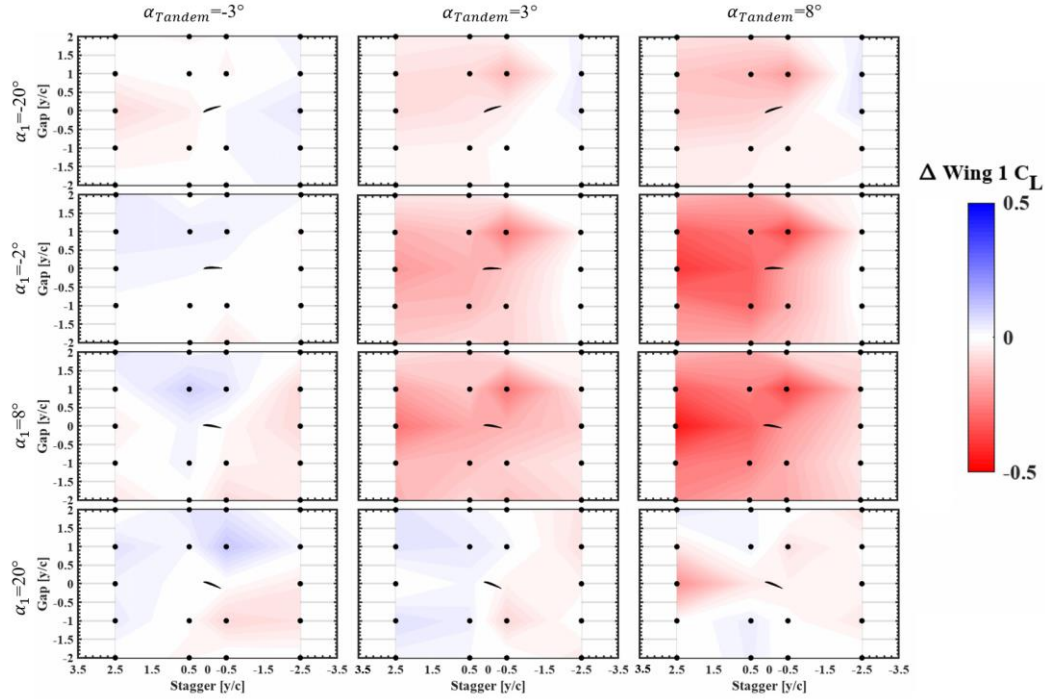


Fig. 10 Contours of Difference in Wing 1 C_L compared to Baseline C_L at different Wing Alpha Combinations

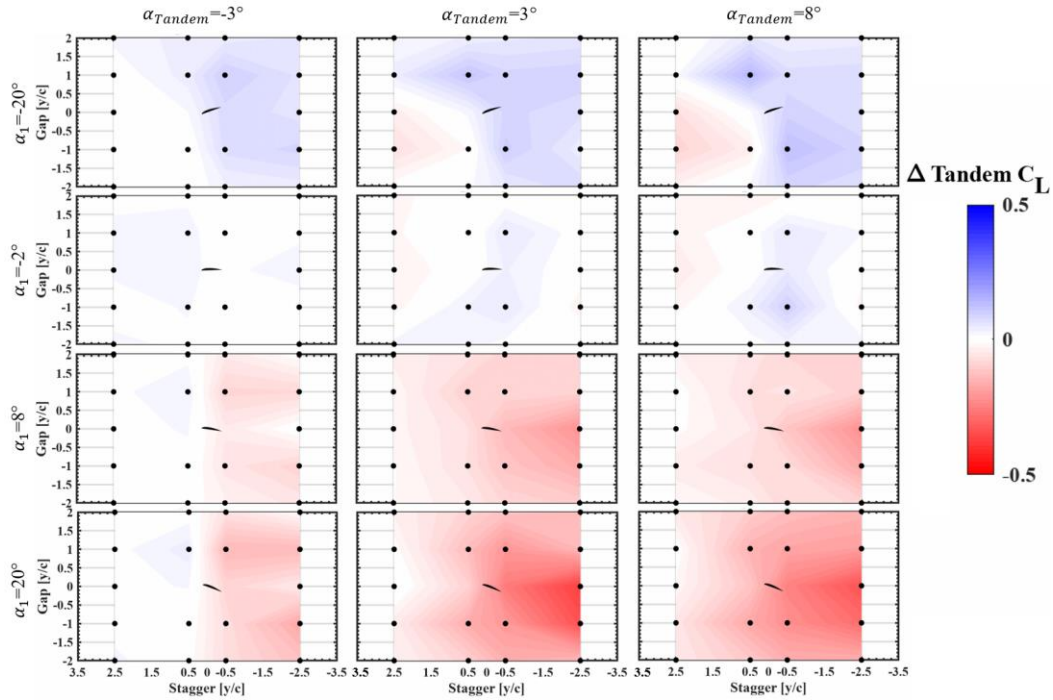


Fig. 11 Contours of Difference in Tandem Group C_L compared to Ideal Two Wing Baseline C_L at different Wing Alpha Combinations

For a comparison between the biplane and tandem configurations, dimension reduction is done with the mean C_L contours in Fig. 12 and Fig. 13 that account for all the interaction effects and the influences of the wings on each other at all Wing 1, Wing 2, and Wing 3 angles of attack at each specific stagger and gap distance. Figure 12 subplots a) and b) compare Wing 1's performance when the biplane or tandem wing configuration is in close proximity. The consequences of the wing downstream of another is evident despite the configuration of the forward wing(s), but it does seem that the biplane configurations span a larger gap width. This is mostly due to the physical configuration having a larger gap footprint compared to the tandems which are stagger dominant. Below in Fig. 13 a) and b), the performance within the two-wing configuration is evaluated. Again, only decrements are present after taking the mean, irrespective of Wing 2 and Wing 3 in a biplane or tandem configuration. The trend of downstream locations suffering decrements is displayed in the mean once more. Comparing the two different two-wing groups shows that the biplane group is slightly more sensitive to the presence of a third wing than the tandem group.

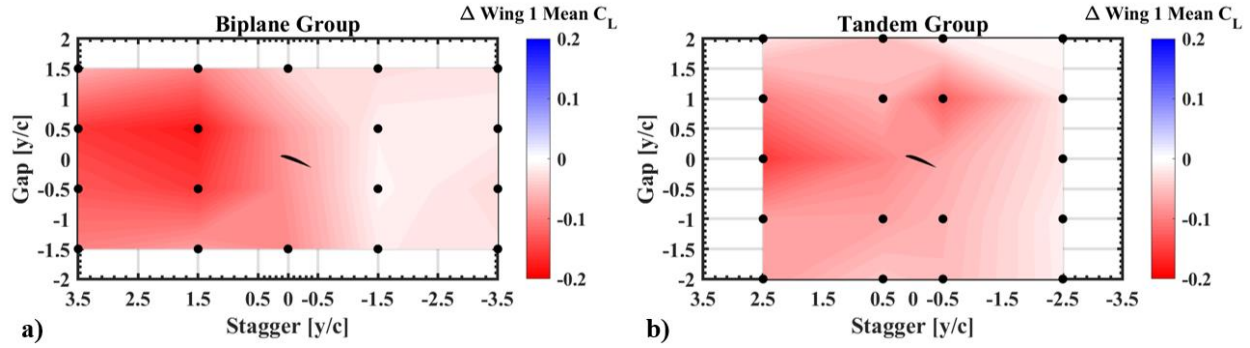


Fig. 12 Difference in Mean C_L of Wing 1 in proximity of a) Biplane Configurations or b) Tandem Configurations from Baseline across all angles of attack at various gap and stagger locations

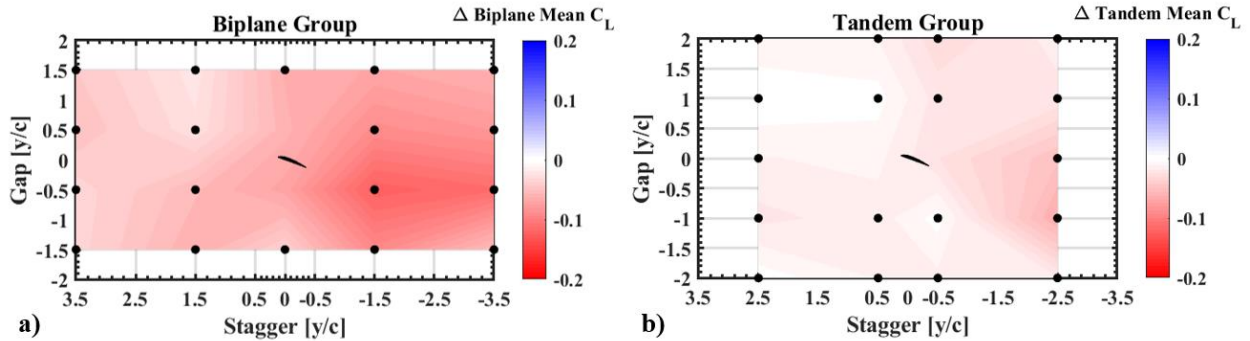


Fig. 13 Difference in Mean C_L of Wing 2 and Wing 3 in a) Biplane Configurations or b) Tandem Configurations from Baseline across all angles of attack at various gap and stagger locations

V. Conclusions

The experimental investigations shown in the current study offer a systematic view to optimize the placements of three wings in stagger, gap, and individual angle of attack. Overall, when multiple lift producing wings are in close proximity, decrements are expected especially on the downstream wings that suffer from the wake of the upstream wings. Possible increments are observed when α_{Stall} is delayed on the downstream wings from downwash effects.

Surprisingly, there did not seem to be a significant difference overall on Wing 1 despite two very different two-wing configurations being placed around it in close proximity. The relative zone of influence on Wing 1 can be seen to be slightly smaller in the tandem tests. While in the two wing studies, the biplane configuration was one of the higher performers, it suffers a larger relative decrement with the presence of a third wing compared to the tandem group, meaning the biplane group is more sensitive to the presence of the third wing when compared to the tandem group. However, the tandem group performed significantly less than the biplane group in terms of system lift

generation. It is also interesting to observe that the areas where Wing 1 is least affected, the biplane and tandem configuration shows the highest performance loss. A system level analysis will reveal the angle of attack ranges of the three wings in consideration that results in a higher system level efficiency which will be quantified in the follow-up paper.

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References

- [1] Sawyer, Nathania. "Zerbe Air Sedan." Encyclopedia of Arkansas, 13 July 2022, <https://encyclopediaofarkansas.net/entries/zerbe-air-sedan-3671/>.
- [2] Faradair BEHA, Retrieved from: <https://www.faradair.com/>
- [3] Prandtl, L., "Induced Drag of Multiplanes" NACA Technical Report TN-182, 1924.
- [4] Munk, M. M., "General Biplane Theory," NACA Technical Report TR-151, 1929.
- [5] Kang, H., Genco, N., Altman, A., "Gap and Stagger Effects on Biplanes with End Plates Part I." AIAA 2009-1085, 47th AIAA Aerospace Sciences Meeting and Exhibit, January 2009, Orlando, Florida.
- [6] Kang, H., Genco, N., Altman, A., "Gap and Stagger Effects on Biplanes with End Plates Part II." AIAA 2009-1086, 47th AIAA Aerospace Sciences Meeting and Exhibit, January 2009, Orlando, Florida.
- [7] Truszkowski, A., Mongin, M., Altman, A., "Feasibility Study on Highly Distributed Lift Configurations." *AIAA SciTech Forum*, 6 Jan. 2019
- [8] Mongin, Michael, et al. "Aerodynamic Feasibility Study on Highly Distributed Lift Configurations Continued." *AIAA 2019 Region III Student Conference*, 6 April 2019
- [9] Mongin, Michael et al. "Extended High Lift Characteristics of Distributed Lift Configurations" *AIAA SciTech Forum*, 23 Jan. 2023
- [10] Jestus, Nevin, et al. "Aerodynamic Characterization of Wing-Wing Interactions for Distributed Lift Application" *AIAA SciTech Forum*, 23 Jan. 2023
- [11] ATI, F/T Gamma Sensor, Retrieved from: https://www.ati-ia.com/products/ft/ft_models.aspx?id=gamma
- [12] PDV, PT-GD201 Rotary Stage, Retrieved from: <https://www.enpdv.com/motorized-stage/motorized-rotation-stage/electric-rotating-machine-motorized-rotation.html>
- [13] ATI, F/T Mini40 Sensor, Retrieved from: https://www.ati-ia.com/products/ft/ft_models.aspx?id=mini40
- [14] Ananda, G., Sukumar, P., Selig, M., "Measured aerodynamic characteristics of wings at low Reynolds numbers." *Aerospace Science and Technology*. 2015.